19. Factors, Multiples & Primes

Year 4 · **Supplementary Worksheet** — extra practice & worked breakdowns

Worked Examples

Example 1 — Listing factors:

List all the factors of 15.

Answer: Find every pair that multiplies to 15: 1×15, 3×5. Factors: **1, 3, 5, 15**.

Example 2 — Listing multiples:

List the first five multiples of 8.

Answer: Keep adding 8: **8, 16, 24, 32, 40**.

Example 3 — Finding common factors:

Find the common factors of 16 and 24.

Answer: Factors of 16: 1, 2, 4, 8, 16. Factors of 24: 1, 2, 3, 4, 6, 8, 12, 24. Common factors: **1, 2, 4, 8**.

Example 4 — Finding the lowest common multiple:

What is the lowest common multiple (smallest shared multiple) of 4 and 6?

Answer: Multiples of 4: 4, 8, 12, 16... Multiples of 6: 6, 12, 18... The smallest one they share is **12**.

Example 5 — Multi-step: lowest common multiple in context:

Two lights flash together at the start. One flashes every 6 seconds, the other every 8 seconds. After how many seconds will they next flash together?

Answer: This is the lowest common multiple of 6 and 8. Multiples of 6: 6, 12, 18, 24... Multiples of 8: 8, 16, 24... They first share 24, so the lights flash together again after **24 seconds**.

Foundation

15 questions

1. Is 6 a prime number or not? Explain briefly.

2. List all the factors of 14.

3. Select ALL of these that are multiples of 6: A) 12 B) 18 C) 20 D) 24 E) 28

4. Is 28 a multiple of 5?

5. What is 6 squared?

6. Is 11 a prime number or not? Explain briefly.

7. Which of these is NOT a factor of 24? A) 1 B) 5 C) 6 D) 8

8. List the first five multiples of 9.

9. Is 45 a multiple of 10?

10. What is 8 squared? A) 16 B) 32 C) 48 D) 64

11. Is 15 a prime number or not? Explain briefly.

12. List all the factors of 18.

13. Which of these is the 4th multiple of 7? A) 21 B) 24 C) 28 D) 35

14. Is 29 a multiple of 4?

15. What is 9 squared? A) 18 B) 72 C) 81 D) 90

Standard

20 questions

1. List all the prime numbers up to 20.

2. Is 16 a square number? What is √16? A) No; not applicable B) Yes; √16 = 2 C) Yes; √16 = 4 D) Yes; √16 = 8

3. Find the common factors of 16 and 20.

4. What is the lowest common multiple (smallest shared multiple) of 4 and 6?

5. List all the factor pairs of 24.

6. Select ALL of these that are prime numbers: A) 37 B) 39 C) 41 D) 44 E) 49

7. Is 64 a square number? What is √64?

8. Find the common factors of 9 and 15.

9. What is the lowest common multiple (smallest shared multiple) of 3 and 4? A) 7 B) 8 C) 12 D) 24

10. List all the factor pairs of 40.

11. List all the prime numbers up to 15.

12. Is 100 a square number? What is √100? A) Yes; √100 = 9 B) Yes; √100 = 10 C) Yes; √100 = 11 D) No

13. Find the common factors of 24 and 36.

14. What is the lowest common multiple (smallest shared multiple) of 4 and 5? A) 9 B) 15 C) 20 D) 25

15. List all the factor pairs of 45.

16. Is 27 a prime number, a square number, or neither?

17. 20 has how many factors in total? List them.

18. Find the highest common factor of 16 and 24. A) 4 B) 8 C) 12 D) 16

19. Is 36 a prime number, a square number, or neither?

20. 30 has 8 factors. Which of these are factors of 30? Select ALL that apply: A) 4 B) 5 C) 6 D) 10 E) 12

Hard

5 questions

1. Two bells ring together at the start. One rings every 9 minutes, the other every 12 minutes. After how many minutes will they next both ring together?

2. 3 and 7 are both prime numbers. What is 3 × 7, and is the answer odd or even?

3. 48 has how many factors in total? List them.

4. Find the highest common factor of 20 and 30. A) 5 B) 10 C) 15 D) 20

5. Is 49 a prime number, a square number, or neither?

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